

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA1219	Course Title:	Computer Architecture
CO Assessed:	CO1 – Precision (BL4)	Assessment:	Assessment Tool 1 – Application based questions
Weightage:	40%	Total Marks:	100
CO1 Statement:	Analyse the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking		
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Section / Batch:	CSA1214 12025	Date:	15/07/26

Instructions to Students

- Identify the relevant computer architecture concepts, functional units, or performance parameters required to solve the given application-based problem.
- Analyse the interaction between CPU, memory, bus structures, and I/O components in the given scenario.
- Apply appropriate instruction execution cycles, addressing modes, and architectural concepts to determine the correct solution.
- Compute performance metrics such as CPI, clock cycles, CPU execution time, speedup, or benchmarking measures using appropriate formulas and calculations.
- Interpret the calculated results and explain their impact on overall computer system performance.

QUESTIONS

1. A web developer notices that the displayed colour does not match the expected output because an incorrect hexadecimal colour code was entered. To identify the source of the mismatch, use an online colour picker or graphics software to compare two hexadecimal colour values (for example, #FF0000 and #F00000). Analyse how changing a hexadecimal value affects the binary representation and the displayed output with suitable illustration.
2. A programmer notices that a program produces different outputs when different input values are used. To examine the effect of varying input values, execute an 8085 program that performs arithmetic operations using different input values. Observe the changes in the Sign, Zero, Carry, and Parity flags after each operation. Analyse how the flag register influences program execution and determines why different inputs produce different flag values.
3. A user notices that a computer responds slowly while editing a document, browsing the internet, and playing audio simultaneously. To analyse the system behaviour, perform these activities together and observe the system's response. Analyse how the CPU, memory, and I/O devices cooperate to execute these tasks, and identify which functional unit is primarily responsible for processing, storing, and receiving data during the operations.

Code of Conduct

I K. Maheshbabu (192525122) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K. Maheshbabu

A Web developer notices that the displayed colour does not match the Expected output because an incorrect hexadecimal colour code was entered. To identify the source of the mismatch, use an online colour picker (or) graphics software to compare two hexadecimal colour values (for example, #FF0000 and #F00000). Analyse how changing a hexadecimal value affects the binary representation and the displayed output with suitable illustration.

Aim:

To Compare two hexadecimal colour codes and observe how a small change in the hexadecimal value affects the binary representation and the displayed colour.

Requirements:-

- Computer / Laptop
- Internet Connection.
- Online Colour Picker (or Graphics Software)

Procedure:-

1. Open an online Colour Picker.
2. Enter the colour code #FF0000
3. Note the RGB values and displayed colour.
4. Enter the colour code #F00000
5. Compare the RGB values and displayed colour.

6. Compare both Context into hexadecimal values into binary.

7. Observe the difference in colour intensity.

Short Explanation:

The colour #FF0000 contains the maximum red value (255), producing a bright red colour. The code #F00000 reduces the red value to 240, making the colour appear slightly darker.

Binary Comparison:

Hexadecimal

#FF0000

#F00000

Binary

11111111 00000000
00000000

11110000 00000000
00000000.

Observation:

- only the red component changes.
- Binary value changes from 11111111 to 11110000
- The displayed colour becomes slightly darker because the red intensity decreases.

Color Palette Generator

space generate lock to keep drag to reorder

Generate

Undo

+ Add Color

Extract

Visualize



#FF0000

Red

Copy

#F00000

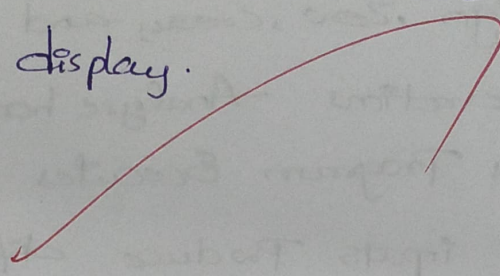
Red

Copy

Result:-

The Experiment shows that even a small change in a decimal (hexa) colour code changes the binary representation and affects the displayed colour.

Therefore, correct hexadecimal values are important for accurate colour display.



2. A programmer notices that a program produces different outputs when different input values are used. To examine the effect of varying input values, execute an 8085 program that performs arithmetic operations using different input values. Observe the changes in the Sign, Zero, Carry and Parity flags after each operation. Analyse how the flag register influences program execution and determine why different inputs produce different flag values.

Aim:-

To study the effect of different input values on the 8085 flag register after performing arithmetic operations.

Requirements:-

- Computer / Laptop.
- 8085 Simulator (GNU Sim8085 / 8085 Simulator)

Procedure:-

1. Open the 8085 Simulator.
2. Write (or) load an arithmetic program (eg., addition).
3. Execute the program with the first set of the input values.

Observe the accumulator and the flag register.

5. observe change the input values and execute the Program again.
6. Compare the values s, z, cy, and p flags.

Example:

Case 1:

INPUT:

• $A = 20H$

• $B = 10H$

Operation:

$$20H + 10H = 30H.$$

Flags:

• $Sign(s) = 0$

• $Zero(z) = 0$

• $Carry(cy) = 0$

• $Parity(p) = 1$

Case (2):

INPUT:

• $A = F0H$

• $B = 20H$

operation:

$$F0H + 20H = 110H.$$

Flags:

- Sign (S) = 0
- Zero (Z) = 0
- Carry (CY) = 1
- Parity (P) = 0

Observations:

INPUT Values	Result	S	Z	CY	P
20H + 10H	30H	0	0	0	1
F0H + 20H	10H	0	0	1	0

Result:

The Experiment shows the different input values produce different arithmetic results and update the flag register accordingly. The Sign, Zero, Carry, and Parity flags indicate the status of the result and influence the execution flow of the 8085 Program.

Registers

A	IE	S	0
BC	14 00	Z	0
DE	00 00	AC	0
HL	00 00	PC	42 0C
PSW	00 00	SP	FF FF
Int-Reg	00	P	1
		C	0

Flag

Load me at

- 1 IDA 2500H
- 2 MOV B,A
- 3 IDA 2501H
- 4 ADD B
- 5 STA 2502H
- 6 HLT
- 7

Decimal - Hex Conversion

Decimal Hex

I/O Ports

-

Memory

-

Data Stack Keypad Memory I/O ports

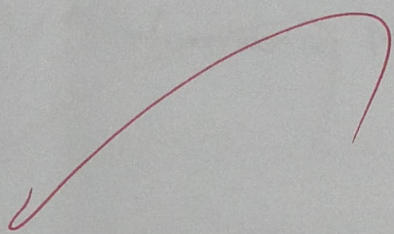
Start 2500h

Address (Hex) Address Data

2500	9472	20
2501	9473	10
2502	9474	30
2503	9475	0
2504	9476	0
2505	9477	0
2506	9478	0
2507	9479	0
2508	9480	0
2509	9481	0
250A	9482	0
250B	9483	0
250C	9484	0
250D	9485	0
250E	9486	0
250F	9487	0

Line No Assembler Message

0 Program assembled successfully



A user notices that a Computer responds a slowly while editing a document, brocosing the internet, and playing audio Simultaneously. To analyse the system behaviour, Perform these activities together and observe the system's response. Analyse how the cpu memory, and I/O devices Cooperate to Execute these tasks, and identify which functional unit is Primarily responsible for Processing Storage, and the recieving data during the operations.

Aim:

To analyse the Performance of the Computer while running multiple applications Simultaneously using Resource Monitor / Performance Monitor.

Requirements:

- Compare / Laptop
- windows Resource monitor (or Performance monitor)
- web Browser.
- MS Word / Notepad
- Audio Player.

Procedure:-

1. Open MSWord (or Notepad) and type Some text
2. Open a web browser and browse any website.
3. play an audio file in the background
4. open Resource Monitor (win+R → type resmon → Enter)
5. Observe CPU, memory, Disk, and Network usage.
6. Record the System Performance and identify the functional units involved.

Observations:-

functional unit

Responsibility.

CPU

Processes instructions and execute tasks.

~~Memory (RAM)~~

Temporarily store data and running programs.

Disk (Storage)

Store files and applications permanently.

I/O devices

Receive input (Keyboard, mouse) and provide output (monitor, speakers).

Aim:

Analysis:-

- The CPU Executes instructions for all running applications.
- RAM Stores active Programs such as the browser, document editor, and audio player.
- The Disk reads applications files and saves documents.
- I/O devices handle user input and audio/video output.
- Running multiple applications increases CPU and memory usage, which may slow down the system.

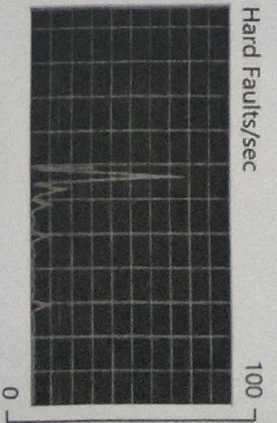
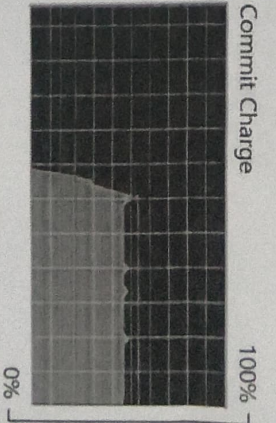
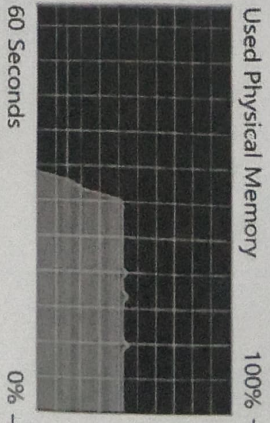
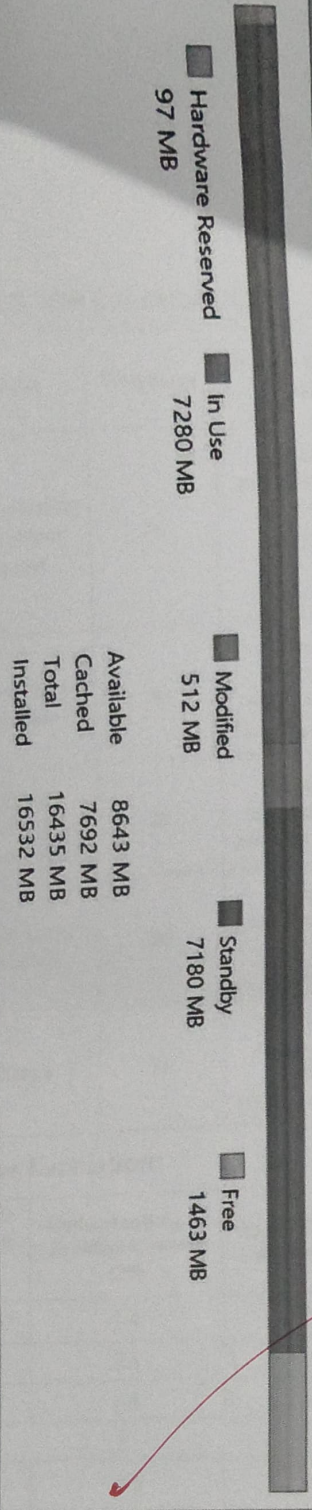
Result:-

The Experiment Shows the CPU, Memory, Disk, and I/O devices work together to execute multiple tasks simultaneously. When many applications run at the same time, system resource usage increases, which can reduce overall performance.

Processes 45% Used Physical Memory

	PID	Hard Faults/sec	Commit (KB)	Working Set (KB)	Shareable (KB)	Private (KB)
☐ Image	11272	4	352,512	198,732	67,220	131,512
☐ msedge.exe	10936	1	124,472	93,456	33,120	60,336
☐ SearchApp.exe	16520	2	98,724	71,884	28,916	42,968
☐ 8085simulator.exe	17328	0	85,632	62,044	22,300	39,744
☐ dwm.exe	10560	1	74,216	55,460	20,256	35,204
☐ explorer.exe	9468	2	66,248	36,808	15,180	21,628
☐ SearchIndexer.exe	912	0	52,640	28,752	11,032	17,720
☐ svchost.exe (LocalService...	11844	1	48,880	26,340	9,940	16,400
☐ OneDrive.exe	11628	0	31,092	19,420	7,512	11,908
☐ ctfmon.exe	11876	0	29,604	17,968	6,540	11,428
☐ Taskmgr.exe						

Physical Memory 7280 MB In Use 8643 MB Available



RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Q. No	Understanding of Problem Context (20)	Application of Concepts (30)	Problem-Solving Approach (20)	Accuracy of Solution (20)	Clarity (10)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	/ 30	/ 20	/ 20	/ 10	/ 100

Course Coordinator

Faculty Incharge